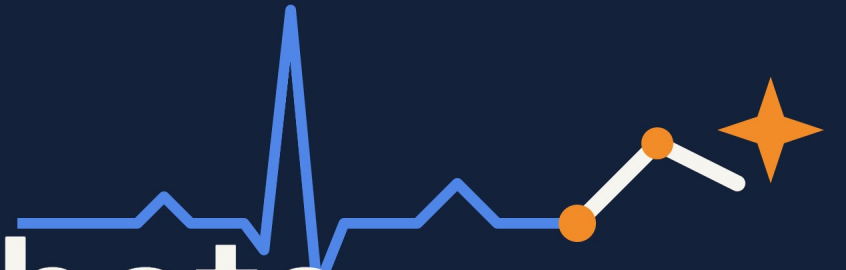


MACHINE LEARNING BOOTCAMP · DAY 5 OF 5

ML on Devices, Robots & Generative AI



Heartbeats, tiny chips, learning robots and machines that create

Soyinka Enoch Oluwatimilehin · 02/10/2026

Quick recap of Day 4

1

Neurons + layers, trained by backpropagation

2

CNNs slide filters over images: 91% on clothes

3

YOLO finds, outlines and poses objects in real time

Warm-up: your smartwatch counts your heartbeat. Where does the “AI” run — on the watch or in the cloud?

Today's plan — 3 hours

Time	Block	What we'll do
0:00 – 0:10	Welcome	Day 4 recap, today's plan
0:10 – 0:51	ML meets the body	ECG signals, heartbeats + Notebook 1
0:51 – 1:25	ML on tiny devices	ESP32, Raspberry Pi, TinyML + Notebook 2
1:25 – 1:35	Break	Stretch, questions
1:35 – 2:14	Robots & RL	Learning by reward + Notebook 3
2:14 – 2:52	Generative AI	How LLMs create text + Notebook 4
2:52 – 3:00	Wrap-up	The 5-day journey, what next

PART 1

ML meets the body

Bioinstrumentation: measuring the human body — and making sense of it.

Signals your body sends



ECG

Heart's electrical activity — rhythm, arrhythmias



EEG

Brain waves — sleep, seizures, brain-computer interfaces



EMG

Muscle activity — prosthetic hands, gesture control



PPG / SpO₂

Light through the skin — pulse and blood oxygen



IMU

Motion sensors — steps, falls, tremor



Temperature

Fever, ovulation tracking, infection alerts

Anatomy of one heartbeat



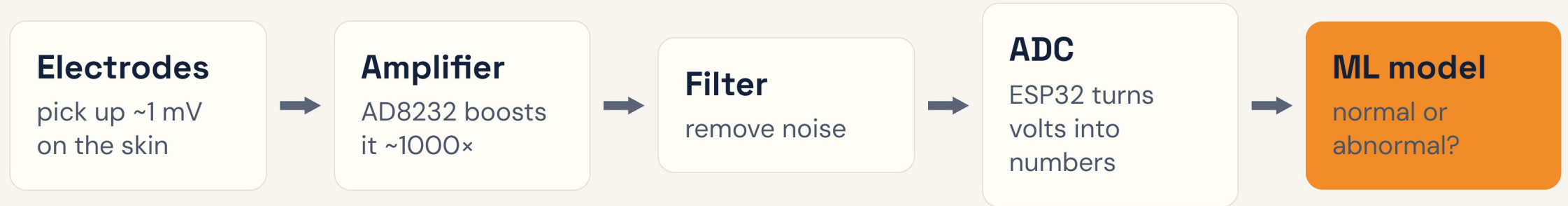
P: top chambers squeeze

QRS: main chambers squeeze — the big spike

T: heart resets

Our recording: **360 numbers per second**

From skin to smart answer



Every stage is engineering you study: circuits, signals and systems, embedded programming — and now ML at the end.

Cleaning the signal: filters

Baseline wander

Slow drift from breathing and movement — below 0.5 Hz

Mains hum

50 Hz from power lines (NEPA!) — 60 Hz in the USA

Muscle noise

Fast wiggles when you move — above ~40 Hz

0.5–40 Hz

Band-pass filter: keep only the heart's frequencies

50 Hz

Notch filter: delete exactly the mains hum

Finding every heartbeat

2,273

heartbeats detected in
30 minutes

2,272

matched the cardiologists'
2,273 labels

75.8

average heart rate (bpm)

Method: filter, then find peaks taller than 0.5 mV and at least 0.25 s apart. Missed **1**, false alarm **1**. Heart-rate variability (SDNN): **48 ms**.

Normal or abnormal heartbeat?

ECG5000: 4,998 beats × 140 samples · 20% held back for testing

99.3%

Logistic regression

99.5%

Random forest

99.3%

1-D CNN · 17,121 weights

Abnormal beats caught (recall): logistic 98.6% · forest 98.8% · CNN 98.8%. In medicine, **a missed abnormal beat is worse than a false alarm.**

ML in healthcare today

Smartwatches

Apple Watch, Fitbit, Samsung detect irregular rhythm (AFib) from the wrist

Medical imaging

AI reads chest X-rays for TB and helps screen for cancer

Microscopy

Count malaria parasites in blood smears

Sleep & stress

EEG, PPG and HRV track sleep stages and stress

Prosthetics

EMG signals control artificial hands

Be careful

Needs regulation, testing on diverse patients, privacy

</> HANDS-ON · 15 MINUTES

Notebook 1

Day5_1_ECG_Signals.ipynb

Part 1 · A real hospital ECG

Part 2 · Filters: remove 50 Hz noise and drift

Part 3 · Find every heartbeat → heart rate

Part 4 · Classify normal vs abnormal beats

PART 2

ML on tiny devices

TinyML: machine learning on a \$5 chip — no internet needed.

Cloud or edge?



Cloud

Huge models, lots of power — but needs internet, has delay, and your data leaves the device



Edge (on the device)

Works offline, answers instantly, keeps data private, runs on a battery for months

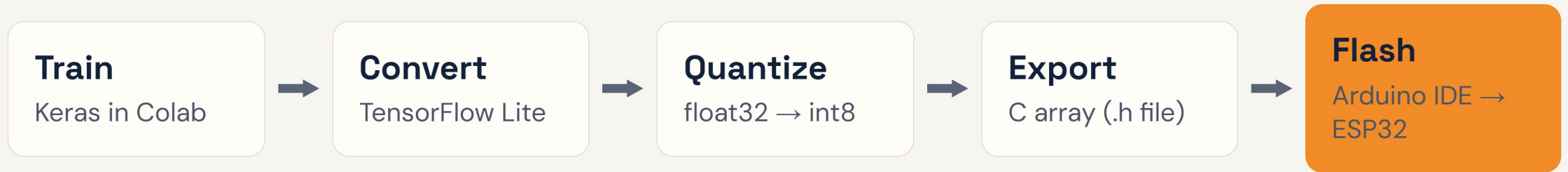
A heart monitor must not stop when the network drops.

Meet the hardware

Device	Memory	Good for
ESP32	520 KB RAM	Sensors + small models, Wi-Fi/Bluetooth, ~\$5
Arduino Nano 33 BLE Sense	256 KB RAM	Built-in mic, IMU — TinyML course board
Raspberry Pi 5	2–16 GB RAM	A full Linux computer: Python, cameras, YOLO
NVIDIA Jetson	4–64 GB	GPU on the edge: robots, drones, video

Your laptop has ~**30,000×** more RAM than an ESP32.

The TinyML workflow



```
converter = tf.lite.TFLiteConverter.from_keras_model(model)
converter.optimizations = [tf.lite.Optimize.DEFAULT]
tflite_model = converter.convert()
```

Quantization: 4 bytes → 1 byte

239.4 KB

Keras file

70.8 KB

TensorFlow Lite float32

23.3 KB

TFLite int8 — accuracy
still 99.3%

Each weight becomes a whole number from -128 to 127: $\text{int8} = \text{round}(\text{value} \div \text{scale}) + \text{zero point}$

Fits the ESP32's 4 MB flash more than 100 times over.

Running it on the ESP32

```
#include "ecg_model.h" // the model as a C array
...
for (int i = 0; i < 140; i++)
    input->data.int8[i] = round(beat[i] / SCALE + ZERO);
interpreter->Invoke();
float p = (output->data.int8[0] - OUT_ZERO) * OUT_SCALE;
if (p > 0.5) Serial.println("Abnormal beat - check!");
```

Sensor: **AD8232** ECG board → ESP32 analog pin → R-peak finder → model → Bluetooth to your phone

Raspberry Pi: a computer in your palm



Same model

Copy `ecg_int8.tflite` and run it with **ai-edge-litert** in Python



Cameras

Run Day 4's YOLO: count people, read plates, spot crop disease



Robots

The “brain” of many student robots — with ROS

```
from ai_edge_litert.interpreter import Interpreter
interp = Interpreter(model_path="ecg_int8.tflite")
```

</> HANDS-ON · 14 MINUTES

Notebook 2

Day5_2_TinyML_ESP32.ipynb

Part 1 · Retrain the heartbeat model

Part 2 · Convert and quantize to int8

Part 3 · Still accurate? How fast?

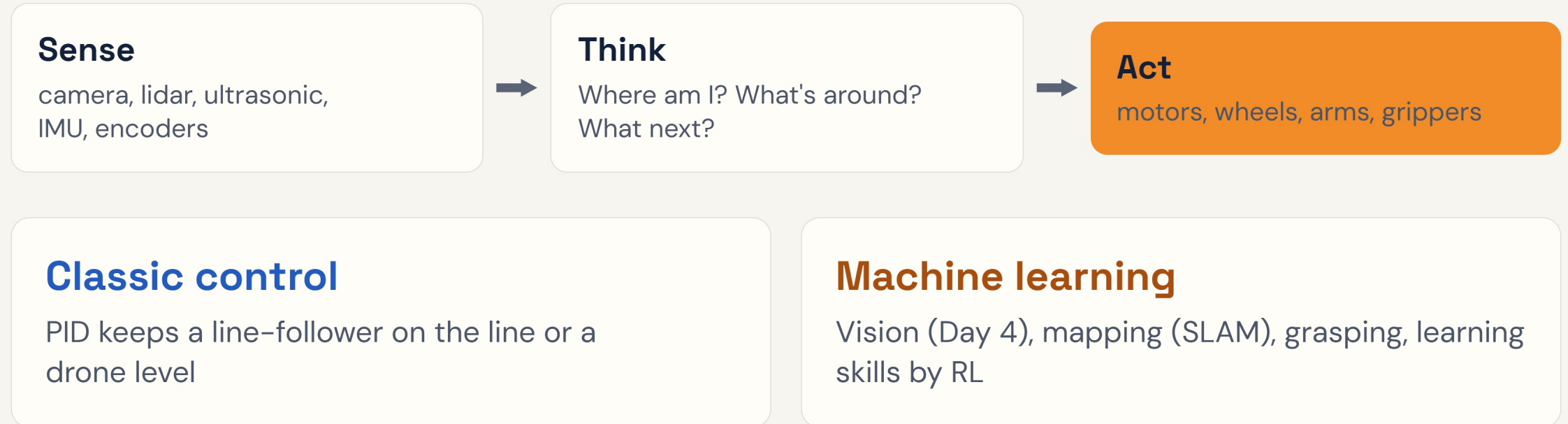
Part 4 · Export a C array + ESP32 sketch

PART 3

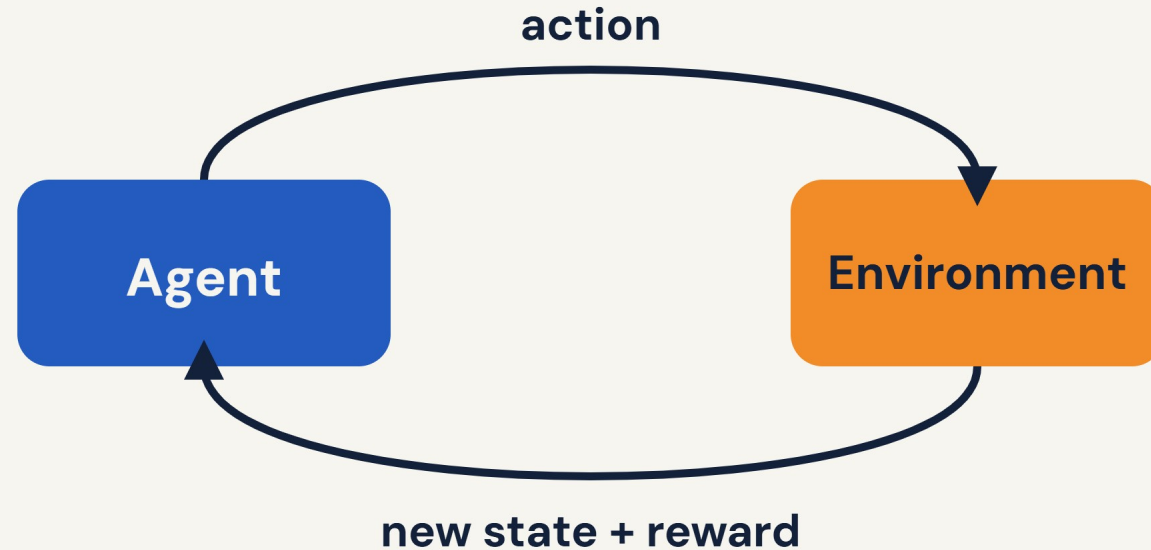
Robots & reinforcement learning

Learning by trial, error and reward.

Every robot: sense → think → act



Reinforcement learning



No right answers given — only **rewards**

Goal: collect the most reward over time

Like training a dog with treats

Q-learning: a table of good moves

$$Q[\text{state}, \text{action}] += \alpha \times (\text{reward} + \gamma \times \text{best } Q \text{ of next state} - Q[\text{state}, \text{action}])$$

Q-table

A score for every (square, move) pair. Starts at zero.

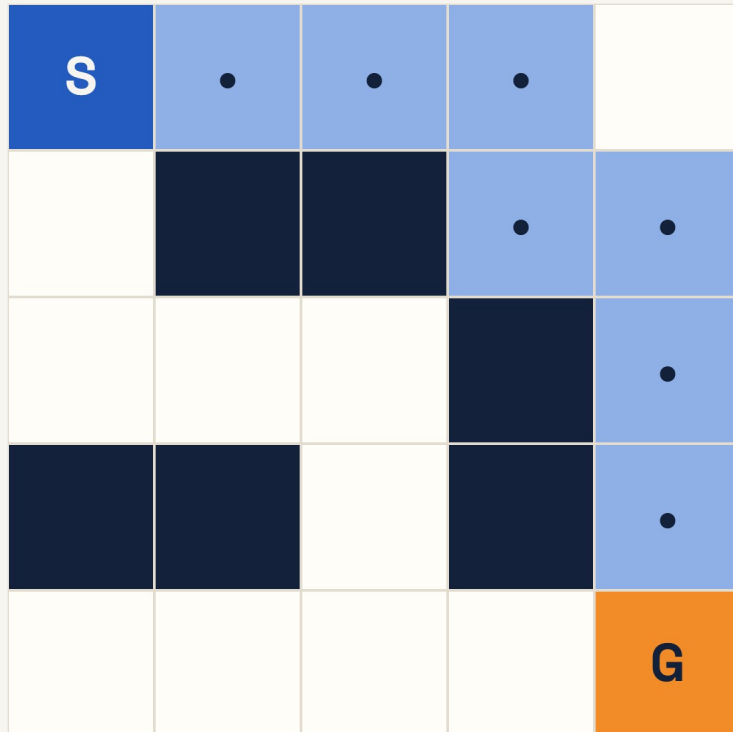
Explore vs exploit

ϵ -greedy: sometimes try a random move, otherwise use the best one

Discount γ

Reward now is worth a bit more than reward later

Warehouse robot: results



Average reward, first 50 tries: **-83**

Average reward, last 50 tries: **+92**

Learned route: **8 moves** — the shortest possible

Nobody told it the way.

Self-driving taxi (Gymnasium)

Pick up a passenger at one of 4 stops, drop them at their destination · 500 states · 6 actions

6/100

Random driver: passengers delivered (avg reward -765)

100/100

Trained driver: 13.4 steps per trip (avg reward +7.6)

5,000 practice trips took 3.9 seconds to learn.

RL in the real world



Games

DeepMind's AlphaGo beat world champion Lee Sedol at Go (2016)



Robots

Legged robots learn to walk and recover from pushes — first in simulation



Chatbots

RLHF: human ratings are the reward that makes ChatGPT and Claude helpful



Energy & traffic

Cooling data centres, timing traffic lights, managing batteries

</> HANDS-ON · 14 MINUTES

Notebook 3

Day5_3_Robots_RL.ipynb

Part 1 · Build a warehouse robot world + Q-learning

Part 2 · See the route, arrows and values it learned

Part 3 · Train a self-driving taxi

PART 4

Generative AI

Machines that create: text, images, code, music.

Predict vs create

Days 1–4: predict

Is this beat abnormal? Which clothing type? What house price?

Generative: create

Write an essay, draw a picture, compose music, generate code

A generative model learns what the data looks like — then produces new examples.

An LLM predicts the next token

The capital of Nigeria is ?



Pick one → add it → predict again
Repeat, token by token, until done

Our tiny model invents names

Trained on 32,033 English names · tokens = 26 letters + “end”

Model	Loss	Invented names
Random letters	3.30	—
Counting (1 letter back)	2.45	reslaynn, siquxxtal, brlyaha
Neural (3 letters back)	2.16	kyeth, juvice, keanna, avorea

Neural model: 11,897 weights · **embeddings** turn each letter into 10 learned numbers

Temperature: safe or creative?

T = 0.5

john, jana, maryn, jaiya, kenzela, mahira

5 of 10 real names

T = 1.0

kyeth, omarib, juvice, zeckrit, keanna, mercelys

9 of 10 new

T = 1.5

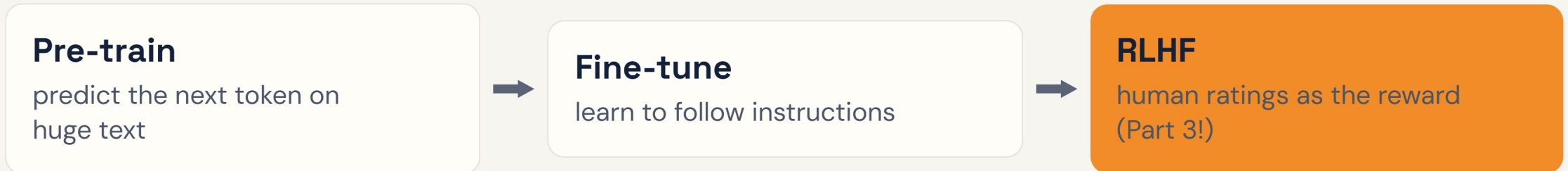
kydvelten, ihanahzyna, khartibremmarlocna

getting strange

Chatbot settings use the same knob: low for facts and code, higher for stories and brainstorming.

From our toy to ChatGPT

Model	Weights	Trained on
Our name model	11,897	32,033 names
GPT-2 (2019)	1.5 billion	~8 million web pages
Today's frontier LLMs	hundreds of billions+	trillions of tokens



The generative AI toolbox



Text & chat

ChatGPT, Claude, Gemini, Llama
— explain, summarise, draft



Code

Write, explain and debug code
— you've seen it in Colab



Images

Diffusion models: Stable
Diffusion, Midjourney, DALL·E



Speech & music

Text-to-speech in local
accents, transcription, music



RAG

Give the model your own
documents so it answers
from them



Agents

Models that use tools: search,
run code, fill forms

Using generative AI well



Do

Give context and examples · ask for steps · check facts and code · use it to learn, not to skip learning



Watch out

Hallucinations (confident but wrong) · bias · private data in prompts · deepfakes · academic honesty



Treat it like a very fast, well-read classmate who is sometimes confidently wrong.

</> HANDS-ON · 13 MINUTES

Notebook 4

Day5_4_GenAI_From_Scratch.ipynb

Part 1 · Counting model: which letter comes next?

Part 2 · A neural language model with embeddings

Part 3 · Temperature: safe vs creative

Part 4 · Colab bonus: talk to a real LLM

Your 5-day journey

Day 1

What ML is, types of learning, the ML workflow

Day 2

NumPy, pandas, EDA, regression & classification

Day 3

Model showdown, tuning, clustering, PCA

Day 4

Neural networks, CNNs, YOLO

Day 5

ECG, TinyML, robots & RL, GenAI

14 notebooks • real data • models you trained yourself.

Where to go next



Build a project

ECG monitor · crop-disease camera · campus parking counter · a chatbot for your course notes



Keep learning

Kaggle (free courses + competitions) · Andrew Ng's ML courses · fast.ai · Hugging Face course · TinyML (Harvard, edX)



Join a community

Data Science Nigeria · Google Developer Groups · Kaggle forums · study groups

Start small, finish it, share it on GitHub and LinkedIn.

Thank you! Questions?

Keep building. Share what you make.

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